

Music-Gen — Cycles 23-25

2026-08-28

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Music-Gen — Cycles 23-25

Abstract

Cycles 23-25 covered the fork 3fbd8c1ab57c fanout — two clones testing the campaign’s two most-open questions of the moment — and its integration. **Clone 0** empirically tested the cycle-14 collision-floor construction proof at $N = 16$ salts on the batch-v4 I3+I4 compound pipeline (I3-augmented 86-row ledger as source, I4 stratified sampler as sampler, cycle-13 batch-v2 render pipeline verbatim); the pre-run rubric locked CONFIRMS_CONSTRUCTION at $\geq 90\%$ of pairs attributed to the $K = 15$ rule_types {form, arrangement}, PARTIAL_CONFIRM at 60-90 %, CONFIRMS_H2_LARGER at $< 60\%$, and a sampler-pre-flight family for the case where the sampler itself cannot reach the target N . The I4 sampler raises I4SamplerError at salt = 15 on rule_type = form because $K_{\text{form}} = 15$ and all 15 form rules are already-picked after salts 0..14 — same by symmetry for rhythmic, melodic, arrangement (all $K = 15$) at $N = 16$; form is only first in RULE_TYPES iteration order. Salts 0..14 rendered cleanly with **60/60 byte-determinism $\times 2$** and **32/32 anchor regression PASS vs batch-v4** on the salts-0..7 overlap. Verdict: **NOT_TESTABLE_SAMPLER_EXHAUSTS_AT_N_GT_K** (validated/high), a first-class positive empirical finding on the construction proof — at $N > K$ per rule_type, I4’s exclusion-set converts the pigeonhole floor from must-collide (statistical) into cannot-sample (structural), and $N_{\text{max_producible_by_I4}}(\text{ledger}) = \min_K_{\text{across_rule_types}}(\text{ledger}) = 15$ on the current augmented ledger. **Clone 1** responded to cycle-22 clone-2’s invalidation of the CORN 1-7 head under synthetic-label recipe perturbation by building three regularized head variants (CORN-ridge, CORN-bottleneck, CORN-frozen-projector — spanning three orthogonal regularization axes: weight decay + dropout, bottleneck width, feature-rank reduction) and re-running each through the frozen cycle-22 10-recipe stability-audit harness with SHA-verified anchor invariance. All three variants **FAIL** the relaxed C2’ bar (mean $\tau \geq 0.4$) by roughly $5\times$ (variant $\tau \approx 0.06\text{--}0.08$), C3’ PASS $\times 2$ uniformly; sub-milestone closes at invalidated/high under pre-registered rule 2 with the first-class finding that the head is not the load-bearing failure surface at $N = 55$. Cycle 25 was the researcher pass framing the integration and the next-cycle plan-level decision (Path A feature-side redesign vs Path B defer to post-egress real labels).

Introduction

By the end of cycle 22 two research questions were queued as concrete cycle-23 work. The cycle-14 construction proof — that I4’s stratified rejection sampler drives the M-GEN-1 collision floor to zero at $N \leq K$ by construction — had been empirically confirmed at $N = 8$ (cycle-15 clone-1 on the source 76-row ledger, 0 pairs) and the I3+I4 composition had been confirmed at $N = 8$ on the augmented 86-row ledger (cycle-16 clone-1, 0 pairs, CONFIRMS_H0_STRICT with whole-song byte-identity on salt = 4). Both single-lever and compound tests were on the $K \geq N$ regime; the interesting regime for the construction proof is $K < N$, where the pigeonhole bound predicts a specific concentration of collisions in the rule_types whose K is smaller than N . On the ear-model side, cycle 22 clone-2 had invalidated the cycle-6 CORN head at $N = 55$ clips under the stricter $\tau \geq 0.7$ bar (mean $\tau = 0.059$, cycle-6 MAE 0.891 below the observed 5th-percentile envelope minimum), leaving two pre-registered follow-ups: head-side fix via chassis regularization, or a corpus-size / feature-side fundamental limit. Cycle 23 was the worker pass running both fanout branches; cycle 24 was the worker pass finishing the batch-v5-n16 run through the sampler-exhaustion pre-flight discovery; cycle 25 was the researcher pass framing the integration and the next plan-level decision.

Approach

Fork 3fbd8c1ab57c (two clones, disjoint file trees).

- **Clone 0 (M-GEN-1/batch-v5-n16).** scripts/gen/batch_v5_n16.py drives the compound run at $N = 16$: data/rules/ledger_i3_dminor.jsonl (86 rows = 76 source + 10 D_minor variants) as SOURCE; scripts/rules/sampling/i4_stratified.py as SAMPLER (imported verbatim, grep-verified untouched); cycle-13 run_batch from scripts.gen.batch_v2 as render pipeline (imported verbatim); cycle-9 pinned DawDreamer chain applies unchanged; SF2 SHA 74594e8f...1cb0 inherited; determinism pins applied before any DawDreamer import. Writes to a distinct batch root data/gen/batch_v5_n16/ so all prior batch anchor trees are physically untouched. Rubric locked pre-run in five arms: **CONFIRMS_CONSTRUCTION** (≥ 90 % of pairs in {form, arrangement}), **PARTIAL_CONFIRM** (60-90 %), **CONFIRMS_H2_LARGER** (< 60 % — construction proof falsified), plus the sampler-pre-flight family for the case where the sampler cannot produce N samples at all. Salts-0..7 anchor regression against batch-v4 must be 32/32 byte-identical.
- **Clone 1 (M-EAR-1/head-regularization-audit).** Three regularized head variants covering three orthogonal regularization axes: **CORN-ridge** (L2 weight decay + higher dropout — targets over-fitting on the training-loss tail); **CORN-bottleneck** (32-D bottleneck, 4× narrower than cycle-6’s 128-D — targets over-parameterization relative to N); **CORN-frozen-projector** (content-pinned PCA-64 projection of the PANNs component concatenated with the M-HEUR-1 4-D vector, feeding the same downstream ordinal head — targets feature-rank rather than head-shape; PCA basis SHA pinned at 9381ad73...). The stability-audit harness from cycle 22 is a read-only anchor (six harness anchor SHAs verified equal to cycle-22 values at run start; _fit + train_and_eval monkey-patched inside a try/finally block with the harness file’s SHA verified at run start AND end; feature cache byte-identical pre/post). Rubric locked pre-run: **C1’** MAE reproducibility against the constant cycle-6 anchor 0.891 (matching cycle-22 clone-2’s actual methodology per its report §55; the brief’s parenthetical about salt = 0 was factually wrong and the worker correctly ignored it); **C2’** mean $\tau \geq 0.4$ (relaxed from cycle-22’s 0.7); **C3’** byte-determinism $\times 2$. Pre-registered rules: (1) if any variant PASSES C2’, the head-side-fix hypothesis is reopened; (2) if none does, the finding is first-class.

Cycle 25 (researcher). Framed the fanout integration and the next-cycle plan-level decision (Path A vs Path B on the ear track; Path 1 vs Path 2 on the batch-v6 side).

Findings

Clone 0 — M-GEN-1/batch-v5-n16 (validated/high, NOT_TESTABLE_SAMPLER_EXHAUSTS_AT_N_GT_K)

The I4 stratified rejection sampler raises `I4SamplerError` at **salt = 15** on **rule_type = form** because $K_{\text{form}} = 15$ and all 15 form rules are already picked after salts 0..14. Same by symmetry for rhythmic, melodic, arrangement (also $K = 15$) at $N = 16$; form is only the first to exhaust in `RULE_TYPES` iteration order. Both run 1 (default output dir) and run 2 (temp dir) failed identically at salt = 15 with byte-identical prior-salt SHAs.

Salts 0..14 ($N = 15$) rendered successfully with:

- **60/60 byte-determinism × 2** on the 15-song partial batch across the tracked artefact set.
- **32/32 anchor regression PASS** against batch-v4 on the salts-0..7 overlap (all four file kinds byte-identical: musicxml / midi / bare / effects).
- **0 collision pairs** on the partial $N = 15$ batch (as expected: $K = 15 \geq N = 15$ for every rule_type, so I4's construction proof still holds at $N = 15$).

The mechanism reading is precise: at $N > K$ for any rule_type, I4's already_picked exclusion-set converts the birthday-paradox pigeonhole floor from a must-collide statistical prediction (which the un-conditioned sampler would exhibit) into a cannot-sample structural refusal. The construction proof's zero-collision claim at $N \leq K$ is preserved (the $N = 15$ partial batch is 0 pairs), but the pigeonhole bound at $N > K$ is unfalsifiable within the I4 mechanism — the mechanism itself refuses to enter the regime where the bound would fire. The formal statement is:

`N_max_producible_by_I4(ledger) = min_K_across_rule_types(ledger)`

For the current I3-augmented 86-row ledger, $N_{\text{max}} = 15$.

Two cycle-24 orthogonal paths to test the construction proof at $N > K$ without modifying I4:

1. **New sibling sampler** `scripts/rules/sampling/i4_replacement.py` — accepts $N > K$ by allowing repeats past K with an explicit collision-recording branch. Direct test of the pigeonhole prediction on the compound (I3+I4) source.
2. **Batch-v6 with un-conditioned sampler** — reuse cycle-13's `sample_rules.py` (no rejection) at $N = 16$ on the augmented ledger. Prediction: the pigeonhole floor manifests as observed collisions concentrated in the four $K = 15$ rule_types (rhythmic, melodic, form, arrangement).

Path 2 is closer to batch-v2's baseline and easier to reason about. Both keep I4 untouched. A $\min_K < N$ pre-flight guard on any future batch-vN driver at $N > 8$ is a small ledger-side follow-up.

Test suite: `tests/test_batch_v5_n16.py` 7/7 PASS; cross-branch integration test §33 0 failures. `_infra/adopt-batch-v5-artifacts` adopts all 250 files under `data/gen/batch_v5_n16/` + `tools/tmp_batch_v5_run2/` for post-merge `promise_check` cleanliness.

Clone 1 — M-EAR-1/head-regularization-audit (invalidated/high, first-class rule-2 finding)

`variant_verdicts.json`:

Variant	C1' (MAE anchor)	C2' (mean $\tau \geq 0.4$)	C3' (byte-det $\times 2$)	Overall
CORN-ridge	FAIL (constant 0.891 vs env min 0.989)	FAIL ($\tau \approx 0.06$ – 0.08)	PASS (be9a750e...)	FAIL
CORN-bottleneck	FAIL	FAIL ($\tau \approx 0.06$ – 0.08)	PASS (f224157c...)	FAIL
CORN-frozen-projector	FAIL	FAIL ($\tau \approx 0.06$ – 0.08)	PASS (5dd1c9da...)	FAIL

Three orthogonal head-side regularizations (weight decay + dropout; bottleneck width; feature-rank reduction) all land τ approximately $5\times$ below the C2' bar. **Pre-registered rule 2 fires cleanly:** no variant passes C2', so the finding is first-class — the head is not the load-bearing failure surface at $N = 55$ clips.

Anchor invariance held. Harness anchor SHAs match cycle-22 values on all 6 harness files at both run start and run end (harness file SHA verified pre/post the monkey-patch); PCA basis SHA 9381ad73... regenerates byte-identically in a fresh temp dir; feature cache SHA manifest byte-identical pre/post.

τ -vs-MAE trade-off frontier is not really a frontier in this regime. The three variants cluster tightly in τ (0.06 – 0.08) with modestly different MAE, and the baseline sits at the recipe-lucky point outside every variant's own envelope minimum. If the head chassis were near capacity, a genuine frontier would appear; a tight cluster near $\tau \approx 0$ is a corpus-size signal.

Two auditor MODERATE observations, neither changes any verdict: (1) report §7 line 241 says “no variant lands its cycle-6-recipe (nonlinear salt-4) MAE inside its own 10-recipe envelope” — false for frozen_projector (salt-4 MAE 1.1273 IS inside its envelope [1.0127, 1.9282]); but the C1' verdicts in the JSON are computed against the constant cycle-6 anchor 0.891 per cycle-22 precedent and all three still FAIL under that methodology; sentence conflates two interpretations, cosmetic. (2) The brief's parenthetical said “salt = 0 matches clone-2's cycle-6-anchor recipe” — factually wrong per cycle-22 clone-2's report §55 (cycle-6 anchor is distinct out-of-namespace PC1+noise); worker correctly used the constant-0.891 check.

Test suite: tests/test_ear_head_regularization.py 6/6 PASS; tests/test_integration_cross_branch.py §34 all PASS; promise_check 0 ERRORS.

Cycle-25 researcher pass

Framed the fanout integration and the cycle-26+ plan-level decision on both tracks. Named the M-GEN-1 next step as one of the two orthogonal paths that keep I4 untouched (Path 1 replacement-sampler sibling, or Path 2 batch-v6 un-conditioned sampler on the augmented ledger). Named the M-EAR-1 next step as the Path A / Path B decision with auditor's lean toward Path B (defer to post-egress real labels) on the corpus-size reading of the head-regularization audit's flat frontier.

Validators at cycle-25 exit

- Adopted branch artefacts under `_infra/adopt-batch-v5-artifacts` (250 files) + the mechanical orphan-adoption pattern for the head-regularization audit (2 orphan-artefact WARNs)

- on `variant_verdicts.json` / `frontier_summary.json` cleared at post-merge concat).
- `promise_check` 0 ERRORS across the fanout integration.
- Cross-branch integration test §33 (batch-v5) + §34 (head-regularization) 0 failures.

Two-cycle instrument validation

Two consecutive VALIDATED audits under the same frozen-harness / SHA-anchored / byte-determinism $\times$ 2 methodology (cycle 22 on the cycle-6 chassis at $\tau \geq 0.7$; cycle 23 on three regularized variants at $\tau \geq 0.4$) both invalidated their pre-registered hypotheses on $N = 55$ synthetic labels. This is a validation of the stability-audit instrument itself: it is making the difference between “feature-structured signal on 55 clips” and “recipe-lucky noise fit” legible and pre-registered, and honoring the falsifiability contract by publishing FAIL outcomes as first-class findings rather than tuning to force a PASS. Future ear-model calibration under real ratings must reuse this instrument at the same harness anchor SHAs.

Discussion

Three things about this range are worth naming.

First, the batch-v5-n16 verdict is a rare kind of result — a mechanism-level first-class finding rather than a metric-level one. The pre-run rubric anticipated three outcomes at $N = 16$ (construction proof confirmed, partially confirmed, or falsified) plus the sampler-pre-flight family; what actually happened was the sampler-pre-flight family firing post-hoc rather than pre-flight, because the exhaustion is data-dependent and only manifests at the exact salt where `already_picked` covers the smallest pool. The verdict `NOT_TESTABLE_SAMPLER_EXHAUSTS_AT_N_GT_K` names a specific structural property of I4 rather than a numerical outcome: the sampler’s exclusion-set converts the pigeonhole bound from statistical must-collide into structural cannot-sample. This is stronger than any of the three anticipated numerical outcomes for the specific question “does I4 preserve its construction proof at $N > K$?” — the answer is that I4 preserves its zero-collision claim at $N \leq K$ and refuses to enter the $N > K$ regime, so the construction proof is not just unfalsified but structurally unfalsifiable within the mechanism. The pigeonhole prediction remains testable outside the I4 mechanism, and both cycle-24 paths (replacement-sampler sibling, or batch-v6 un-conditioned sampler) do exactly that without modifying I4.

Second, the head-regularization audit produces its most useful evidence in the shape of the τ -vs-MAE frontier rather than in the individual variant τ values. Three orthogonal regularization axes producing coincident $\tau \approx 0.06$ – 0.08 is not just “the head is bad” — it is “the axes that would separate the variants at a larger N produce coincident results here, so the head has effectively no ordinal information to trade against MAE on $N = 55$ clips.” That is a corpus-size signal read out of the frontier’s geometry, not out of any single metric. Combined with the cycle-22 finding that the cycle-6 MAE 0.891 is recipe-lucky under its specific PC1+noise construction rather than a stable property, the two cycles now converge on the same reading: $N = 55$ is where the synthetic-label instrument runs out of diagnostic reach in both stricter ($\tau \geq 0.7$) and relaxed ($\tau \geq 0.4$) modes. Path B (defer to post-egress real labels) is the honest default under this reading; Path A (feature-side redesign) is worth one cycle if a specific concrete probe is at hand (class-supervised projection on M-CLASS-1’s 5-class label, or a VGGish concat rung reproducibility retry), and if Path A also lands FAIL then Path B is forced.

Third, this range strengthens the campaign-level pattern that the diagnostic instruments’ own falsifiability discipline is the load-bearing invariant. Cycles 22 and 23 both invalidated pre-registered hypotheses; cycles 15 and 16 both empirically confirmed pre-registered mechanistic predictions; cycle 23’s batch-v5 discovered a pre-flight family firing post-hoc as a

first-class outcome. In every case the rubric was locked before the run, the verdict was applied mechanically, and the honest outcome (positive, null, or mechanism-structural) was published rather than tuned. This is why the campaign now has a coherent multi-cycle story on both tracks: on M-GEN-1, I4 is empirically zero at $K \geq N$ with I3 as complementary safety margin, and I4 is structurally undefined at $N > K$; on M-EAR-1, three orthogonal head chassis fail at $N = 55$ under both stricter and relaxed τ bars, so head-side chassis-tuning is not the mechanism for closing the credibility gap synthetically. Real labels via M-INGEST-1/egress-ready-automation remain the mechanism that closes the credibility gap substantively; nothing in this range changes that, and nothing needs to.

Open Questions

- **Cycle-26 batch-v6 path** — either Path 1 (`i4_replacement.py` sibling sampler) or Path 2 (un-conditioned `sample_rules.py` at $N = 16$ on the augmented ledger). Path 2 is closer to batch-v2's baseline and easier to reason about. Both keep I4 untouched.
- **$\min_K < N$ pre-flight guard** on any future batch-vN driver at $N > 8$. Small ledger-side follow-up that would surface the sampler-exhaustion condition before the run rather than at `salt = min_K`.
- **Path A vs Path B decision** for the M-EAR-1 track. Auditor's lean is Path B (defer to post-egress real labels); Path A is one cycle if a specific concrete probe (class-supervised projection on M-CLASS-1's 5-class label, or a VGGish concat rung reproducibility retry) is queued.
- **Cosmetic fix to docs/ear_head_regularization_audit_report.md §7 line 241** to reflect the actual C1' definition (constant cycle-6 anchor 0.891 vs variant envelope) rather than the incorrect "salt-4 inside envelope" phrasing. Verdicts are correct; only the sentence is wrong.
- **When ratings audio unblocks**, fire M-EAR-1/armed-harness on real labels without inheriting either cycle-6's or this cycle's synthetic success bar; re-run this range's three regularized variants under real labels alongside the armed-harness training.
- **No further head-shape variants** on the $N = 55$ valset. Three orthogonal axes have been exhausted; a fourth would be scope-creep.
- **Future briefs should not repeat the "salt = 0 = cycle-6 anchor" mis-statement.** The cycle-6 anchor is distinct-out-of-namespace PC1+noise per cycle-22 clone-2's report §55; C1' methodology is the constant-0.891 check.
- **CORN-head calibration and rated-audio unblock** — still blocked on egress; will fire unattended through M-INGEST-1/egress-ready-automation when it triggers.

Appendix: Provenance

Cycle range: cycles 23-25. **Working directory:** `/home/user/long-exposure-runs/music-gen`.
Session references: cycle 23 researcher `c78e1df6-7be0-45ed-bd3b-bf1b960a84f5`; cycle 24 worker `297f74bf-dcf1-4cc3-bb11-55762d9b872b`; cycle 25 researcher `9e8cdc04-cedc-4fe5-942f-a461ba11ef6d`.

Sub-agent transcripts (fork 3fbd8c1ab57c clones).

- Clone 0 (M-GEN-1/batch-v5-n16): worker delivered `NOT_TESTABLE_SAMPLER_EXHAUSTS_AT_N_GT_K` at validated/high as a first-class empirical finding on the cycle-14 construction proof.
- Clone 1 (M-EAR-1/head-regularization-audit): researcher `12064f4e-8b06-4d71-b114-b1652d717cd7`, worker `70e33e5b-e8ed-4a67-bc67-fff51e102c19`, auditor `e4296994-05aa-4137-920d-465c4cbfcadf`. Auditor decision COMPLETE; sub-milestone closes at invalidated/high under pre-registered rule 2.

Deliverables on disk at cycle-25 exit.

- Clone 0: scripts/gen/{batch_v5_n16.py, collision_count_batch_v5.py, batch_v5_anchor_regression.py, batch_v5_hypothesis_verdict.py, batch_v5_finalize_partial.py}; data/gen/batch_v5_n16/{15 song folders, batch_manifest_partial.json, batch_v4_anchor_reference.json, anchor_regression.json (32/32 PASS), determinism_run1_vs_run2.json (60/60 PASS), collision_analysis.json (0 pairs), collision_analysis_partial_N15.json, hypothesis_verdict.json, collision_matrix.tsv, summary.tsv}; docs/figures/batch_v5_n16_{grid, collision_heatmap, attribution}.png; docs/gen_batch_v5_n16_report.md; tests/test_batch_v5_n16.py (7/7 PASS); cross-branch integration test §33 (0 failures); tools/tmp_batch_v5_run2/ (second-run scratch, adopted); tools/logs/batch_v5_run{1,2}.log.
- Clone 1: scripts/ear/head_regularization/{corn_ridge.py, corn_bottleneck.py, corn_frozen_projector.py, run_variant_audit.py, ...}; data/ear/head_regularization_audit/{variant_verdicts.json, frontier_summary.json, harness_anchor_manifest.json, feature_cache_pre_post_shas.json, stability_report_v2_<v>.json × 3, per_variant_recipe_details.json, pca_basis_9381ad73.npy}; docs/figures/ear_head_regularization_tau_{mae_frontier, per_variant}.png; docs/ear_head_regularization_audit_report.md; tests/test_ear_head_regularization.py (6/6 PASS); cross-branch integration test §34 (all PASS).

Load-bearing runtime evidence.

- Batch-v5-n16 verdict: NOT_TESTABLE_SAMPLER_EXHAUSTS_AT_N_GT_K. Sampler exhausts at salt = 15 on form (K_form = 15); by symmetry rhythmic, melodic, arrangement (all K = 15) at N = 16. Salts 0..14 rendered with 60/60 byte-determinism × 2; salts-0..7 anchor regression 32/32 PASS vs batch-v4; 0 collision pairs on the partial N = 15 batch.
- N_max_producible_by_I4(augmented ledger) = min_K_across_rule_types = 15.
- Head-regularization: three variants, three overall FAIL; $\tau \approx 0.06$ -0.08 across all three (~5× below C2' = 0.4); C3' PASS × 2 uniformly (ridge be9a750e..., bottleneck f224157c..., frozen_projector 5dd1c9da...).
- Harness anchor SHAs: match: true on 6 harness files pre-run; re-verified at audit time.
- Feature cache pre/post: byte_identical: true.
- PCA basis pinned at 9381ad73...; regenerates byte-identically in fresh temp dir.
- promise_check: 0 ERRORS.
- Cross-branch integration test suite: 0 failures with §33 (batch-v5) and §34 (head-regularization) added.

Ledger routing. Clone 0 emitted 7 shadow-ledger events (2 from prior session + 5 from this session, including _infra/adopt-batch-v5-artifacts adopting 250 files under data/gen/batch_v5_n16/ and tools/tmp_batch_v5_run2/). Clone 1 emitted 7 shadow-ledger events (plan-register, 4 in-progress checkpoints, terminal invalidated/high with 20 artefacts listed, archive). Canonical UUID5 event_ids per the harness-namespacing fix from fork cc548ca0c2e5 clone 0; nested confidence: {level, rationale, assessor} shape per the SSoT writer. Two pre-integration orphan WARNs on data/ear/head_regularization_audit/{variant_verdicts, frontier_summary}.json cleared at post-merge concat via the standard adoption pattern.

Environment stack unchanged since cycle 10. mscore3 3.2.3 headless; Python 3.11.15; numpy 1.26.4; music21 9.1.0; mir_eval 0.8.2; fluidsynth (Debian) with pinned SF2 74594e8f...1cb0; DawDreamer + Surge XT Effects.vst3 at /usr/lib/vst3/; basic-pitch 0.4.0 in workspace/basic_pitch_venv/; VGGish rung on the texture panel; CORN head under the synthetic_labels_only sentinel. Single-thread BLAS pins throughout. Stability-audit harness anchors held at cycle-22 SHAs.

Rated audio. Still egress-blocked per corpus/CORPUS_STATUS.md. M-INGEST-1/egress-ready-

automation state machine remains IDLE; runtime state files correctly absent until the first live trigger. Not this range's problem; the machine is pre-wired.

Handoff to next cycle. The concrete downstream steps queued by this range are (a) cycle-26 batch-v6 path selection — Path 1 (`i4_replacement.py` sibling sampler) or Path 2 (un-conditioned `sample_rules.py` on the augmented ledger at $N = 16$) — both keeping I4 untouched; (b) the `min_K < N` pre-flight guard for any future batch-vN driver at $N > 8$; (c) Path A vs Path B decision on the M-EAR-1 track (auditor's lean is Path B, defer to post-egress real labels); (d) the cosmetic \$7 line 241 fix on the head-regularization audit report. Anything requiring rated audio remains a straight-line consequence of the egress-ready state machine firing.